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Decision letter (Initial Submission)

Human-in-the-Loop Automation for Rapid Dataset Building in Field-Deployed Intelligent Systems

Subject Decision on T-ASE-2026-1694 - Rejection

Date sent 1 June 2026 at 14:01 GMT-7

From r.hytowitz@ieee.org

To mwmaster@ucsc.edu

CC [Show 7 addresses](#)

Dear Mr. Masters:

This e-mail concerns your manuscript T-ASE-2026-1694 Human-in-the-Loop Automation for Rapid Dataset Building in Field-Deployed Intelligent Systems.

The review process is now complete. Your paper was reviewed, and I have included those reviews below. On the basis of the reviewers' ratings and comments, I regret to inform you that your paper cannot be published in IEEE Transactions on Automation Science and Engineering.

The reviewers agreed that the paper in its current form lacks in several aspects and therefore is not suitable for publication.

Please check your Author Center to view any attachments that the reviewers may have uploaded.

We thank you for your interest in the IEEE Transactions on Automation Science and Engineering and hope that you will consider submitting future papers here. If you have any related concerns, do not hesitate to contact me.

Sincerely,

Prof. Zhijun (2026) Li

* * * * *

Associate Editor

Comments to the Author:

This paper discusses Human-in-the-Loop Automation for Rapid Dataset Building in Field-Deployed Intelligent Systems. Currently, several critical problems exist.

1. The whole paper reads as a system engineering implementation. However, some details, such as the parameter settings, are not provided, which weakens reproducibility.
2. The reviewers and I are both concerned about the algorithm description and its strong dependence on restrictive geometric assumptions. Please refer to the reviewers' files for more detailed comments.
3. In the experimental validation parts, please consider broader field evaluations, stronger baselines, and deeper sensitivity analyses of calibration, pose uncertainty, terrain slope, and annotation parameters.

Reviewer's comments to author:

Reviewer: 1

Comments to the Authors

Please comment on Major Contribution of the Paper, Organization and Style, Technical Accuracy, Presentation and Adequacy of Citations (List possible additions if needed).

The manuscript addresses the challenge of high-cost image-level annotation and rapid domain-specific dataset construction for field-deployed intelligent perception systems. It proposes a human-in-the-loop UAV-based georeferenced annotation framework, termed BirdsEye, which places RTK geotags in the physical world and projects them into UAV imagery to generate training labels, thereby reducing repetitive frame-by-frame manual annotation. The paper also derives a first-order pose-to-pixel uncertainty propagation model and evaluates the system through Monte Carlo simulations and a case study on agricultural pest burrow detection. Overall, the work has clear engineering relevance and presents a relatively complete pipeline covering field data collection, georeferenced label generation, model training, and georeferenced detection. However, the theoretical novelty and experimental evidence are not yet sufficient to support the level of contribution claimed in the manuscript. In its current form, the paper is closer to a useful system integration and case-study validation than to a work that makes a substantial advance in automation theory, robotic learning, or perception modeling. Therefore, I cannot recommend publication in its present form. My major concerns are as follows.

1. The proposed framework is mainly built upon established components, including RTK-based geospatial tagging, camera projection, direct georeferencing, SE(3)-based pose perturbation modeling, and CNN heatmap training. While integrating these components into a UAV-based field dataset construction workflow is practically useful, the manuscript does not clearly demonstrate a substantive advance over existing georeferenced annotation methods, robot-assisted annotation, AR/marker-based annotation, semi-automatic labeling, or modern interactive annotation tools. The claimed "human-in-the-loop automation architecture" and "closed-form uncertainty propagation" are better characterized as system integration and standard error-propagation analysis rather than new automation theory, learning methodology, or robotic decision-making.
2. The paper presents first-order covariance propagation from pose uncertainty to pixel-space error as an important theoretical contribution. However, this analysis mainly relies on standard Taylor linearization, Jacobian-based uncertainty propagation, and Monte Carlo validation, which limits its theoretical depth. More importantly, the derivation of the maximum allowable pose uncertainty appears to suffer from notational ambiguity and a potential inequality-direction

issue. If the goal is to satisfy a prescribed pixel-precision constraint, the sensor pose uncertainty should be upper-bounded, rather than expressed in a form resembling a lower-bound condition. This issue directly weakens the credibility of the claimed “operational bound” and “validity regime characterization,” and it also reduces the practical value of the analysis for system design.

3. The main efficiency comparison is conducted against GUI-based frame-by-frame annotation performed by 15 non-specialist annotators. This baseline does not adequately represent currently available high-efficiency annotation workflows. The manuscript does not compare BirdsEye with professional annotation platforms, active learning pipelines, semi-supervised annotation, foundation-model-assisted labeling, SAM-like interactive segmentation, or orthomosaic-based geospatial labeling. As a result, the reported “22–38× per-worker productivity” and “3–5× throughput” improvements are valid only under a specific weak baseline and particular accounting assumptions. They do not yet establish a clear advantage over current mainstream or state-of-the-art dataset construction tools.

4. On the independent test site, the model achieves a recall of 83.7%, but only 35.1% of the detection clusters are matched to human geotags. This suggests a potentially high false-positive rate. The authors argue that many unmatched detections may correspond to real targets that were not labeled during ground-truth collection. While this explanation is plausible in field conditions, it is not supported by independent expert verification, secondary annotation, or blinded review. Therefore, the current results cannot rule out the possibility that the model produces a large number of false positives. For a paper emphasizing reliable and rapid dataset construction for field-deployed intelligent systems, the presented performance evidence is not sufficiently convincing.

5. BirdsEye depends on several restrictive assumptions: the target objects must be relatively static in geospatial coordinates; RTK positioning must remain stable; UAV pose estimation and camera calibration must be reliable; the visible terrain within a frame must be approximately planar; and target height or local terrain variation must not introduce substantial projection errors. These assumptions may hold for pest burrows in relatively flat agricultural fields, but they may not hold in uneven terrain, heavily occluded scenes, dynamic-object settings, 3D structural environments, dense vegetation, or complex industrial sites. Although the manuscript includes a slope sensitivity analysis, its broader claims still tend to generalize the method to field robotics and outdoor intelligent systems more broadly than the current evidence supports.

6. The core empirical validation is centered on agricultural pest burrow detection, with a relatively concentrated target type, scene structure, and acquisition condition. This case study demonstrates feasibility for one specific application, but it is insufficient to justify the method as a general framework for field perception dataset construction. To strengthen the paper, the authors should provide evidence on at least one additional target category, terrain condition, flight altitude/speed setting, or sensor configuration. A more systematic sensitivity analysis with respect to RTK error, camera calibration error, annotation density, soft-label width, heatmap thresholding, and DBSCAN parameters would also be necessary.

7. The manuscript uses several different productivity expressions, including “3–5×,” “300%–500% increase,” and “22–38× per-worker productivity,” which may confuse readers about the actual magnitude and accounting basis of the reported improvement. The authors should consistently report metrics such as frames per worker-hour, valid labels per worker-hour, total workflow time, and whether flight time, field annotation, post-processing, and human verification are included. In addition, the current contribution list places the system architecture, theoretical derivation, Monte Carlo validation, and validity-regime characterization at the same level, which makes the contributions appear scattered and somewhat overstated. A more appropriate positioning would be a deployable georeferenced annotation system and its empirical evaluation for a specific class of field applications, rather than a broadly general theoretical breakthrough.

8. Some expressions are relatively informal for an archival journal paper, such as “The catch is that” and “What is more.” These should be replaced with more formal academic wording. In addition, some system diagrams and workflow descriptions are information-dense. The paper would benefit from a clearer separation among the label-generation stage, the model-training stage, and the georeferenced inference stage, which would improve the readability and logical

flow of the system description.

Reviewer: 2

Comments to the Authors

Please comment on Major Contribution of the Paper, Organization and Style, Technical Accuracy, Presentation and Adequacy of Citations (List possible additions if needed).

This paper presents **BirdsEye**, a UAV-based human-in-the-loop annotation framework for rapid dataset generation in field robotics. The core idea is to replace frame-by-frame image labeling with georeferenced field annotations. Human operators place RTK-GPS geotags on target objects, and UAV imagery is later automatically annotated by projecting these spatial labels into image space. The paper also derives a first-order analytical model for propagating platform pose uncertainty into pixel-space annotation uncertainty and validates this model against Monte Carlo simulations. A case study on agricultural pest burrow detection is used to demonstrate annotation throughput gains.

The problem is practically relevant, and the system has engineering value. However, the paper is not sufficiently strong as a research contribution. The method is largely an integration of direct georeferencing, camera projection, RTK-based field annotation, and standard detector training. The technical novelty is limited, the experimental validation is narrow, and the claimed productivity and detection benefits are not rigorously established.

Major Concerns

1. Limited methodological novelty

The main annotation pipeline is based on direct georeferencing: RTK-GPS field labels are projected into UAV images using calibrated camera geometry and pose estimates. This is a reasonable engineering design, but it is not a fundamentally new annotation or learning method. The paper does not clearly distinguish its contribution from existing geospatial annotation, UAV mapping, and direct georeferencing systems.

The uncertainty analysis is also based on standard first-order covariance propagation through a camera projection model. While useful, the derivation follows conventional SE(3) perturbation and Jacobian-based uncertainty propagation. The paper should clarify what is novel beyond applying known tools to this specific annotation workflow.

2. Strong dependence on restrictive geometric assumptions

The annotation pipeline assumes that the visible terrain in a frame can be approximated by a local plane and that depth can be represented using a radar altimeter measurement. This assumption is central to the projection process. Although the authors analyze terrain slope effects, the system remains limited to relatively flat agricultural fields. The method may degrade substantially on uneven terrain, vegetation-rich areas, curved surfaces, terraced fields, or cluttered outdoor environments.

This weakens the paper's broader framing around "field-deployed intelligent systems," since the demonstrated operating regime is much narrower than the title and motivation suggest.

3. Productivity claims are not rigorously controlled

The paper reports large annotation productivity gains, including 3–5× throughput improvement and 22–38× worker-yield improvement. However, the comparison against manual annotation is

not sufficiently controlled. The manual baseline uses non-specialist annotators labeling random frames, whereas BirdsEye uses field geotagging plus automated projection. These workflows differ substantially in task definition, worker expertise, annotation target visibility, and quality-control requirements.

Moreover, the paper notes that productivity can be increased arbitrarily by increasing camera frame rate. This makes the reported productivity metric partly dependent on acquisition settings rather than actual human annotation efficiency. A fairer comparison should normalize for annotation quality, visible object count, object ambiguity, and target recall.

4. Detection results are weak and difficult to interpret

The downstream pest-burrow detector achieves 83.7% recall but only 35.1% precision on the unseen test site. The paper argues that many unmatched detections may correspond to real but unlabeled targets, but this claim is not rigorously verified. If the ground-truth set is incomplete, then the evaluation protocol is not reliable; if it is complete, then the precision is very poor.

In either case, the detection experiment does not convincingly demonstrate that the generated dataset supports accurate perception model training. The paper should include a more reliable test set with exhaustive expert validation, stronger baselines, and standard detection metrics.

5. Limited experimental scope

The experimental case study is restricted to pest-burrow detection in a small number of agricultural sites. This is insufficient to support broad claims about field robotics, environmental monitoring, infrastructure inspection, or general dataset-building workflows.

The system is not evaluated across diverse object classes, terrain types, flight heights, lighting conditions, camera configurations, or sensor noise profiles. The paper also lacks comparison with alternative UAV annotation strategies, such as orthomosaic-based annotation, active learning, semi-supervised labeling, or existing geospatial labeling tools.

6. Reproducibility and implementation details remain incomplete

Several practical details are under-specified, including the robustness of time synchronization, calibration drift, geotag placement accuracy, frame selection, label radius selection, DBSCAN parameter sensitivity, and the effect of soft-label width on training. The method is highly dependent on these implementation choices, but the paper does not provide sufficient ablation or sensitivity analysis.

Minor Concerns

1. The title and abstract overstate the generality of the approach.
2. The open-source link appears anonymized or placeholder-like, which weakens reproducibility at review time.
3. The detector architecture is basic and not central to the paper, but the paper still uses it to support practical impact claims.
4. Some evaluation metrics are nonstandard and should be better justified.
5. The ethical discussion is appropriate but does not compensate for the limited technical validation.

Reviewer: 3

Comments to the Authors

Please comment on Major Contribution of the Paper, Organization and Style, Technical Accuracy, Presentation and Adequacy of Citations (List possible additions if needed).

This manuscript presents BirdsEye, a rapid dataset construction framework for field-deployed UAV perception systems. The authors integrate RTK ground geotags, UAV direct georeferencing, and image-label generation into a complete human-in-the-loop dataset construction system. The engineering implementation is relatively complete. To address the annotation bottleneck for small-object UAV imagery in field environments, the paper proposes a workflow in which objects are labeled once in spatial coordinates and then automatically projected into multiple image frames. This is practically useful. The paper also derives an analytical covariance propagation model from SE(3) pose perturbations to pixel-space uncertainty and discusses the validity regime of the linearized model.

Overall, this is a system paper with a relatively complete engineering implementation. However, from the perspectives of novelty and rigor of validation, the current version still has clear weaknesses. The core method is essentially based on RTK geotags, direct georeferencing, and image projection. The idea is reasonable, but not highly original. The uncertainty propagation derivation is also a standard first-order covariance propagation technique in robotic vision. Therefore, the main added value of the paper lies in system integration and application validation. More importantly, several key claims regarding annotation efficiency, detection performance, and the uncertainty model appear to be overstated or insufficiently validated.

The paper does not answer several important questions. Does a model trained with BirdsEye-generated labels perform comparably to one trained with manually annotated bounding boxes or point labels? Are soft labels actually better than hard labels? If the authors claim that uncertainty analysis can guide the choice of soft-label width, is this strategy better than using a fixed sigma? How does the system perform under different flight altitudes, camera poses, terrain slopes, and degraded RTK accuracy? How does it compare with annotation after orthomosaic generation, conventional GIS-based annotation, active learning, or semi-supervised annotation tools? The absence of these baselines and ablation studies makes the claimed superiority of the method remain mostly at the level of system demonstration. I recommend adding stronger baseline comparisons and ablation experiments.

The derivation of the operational bound in the uncertainty propagation section appears mathematically inconsistent. Starting from $\Sigma_u = J_{\xi} \Sigma_0 J_{\xi}^T$, the appropriate spectral norm bound is $\|\Sigma_u\|_2 \leq \|J_{\xi}\|_2^2 \|\Sigma_0\|_2$. However, the manuscript then concludes that $\lambda_{\max}\{\Sigma_u\} \geq \lambda_{\max}\{\Sigma_0\}^2$, which is inconsistent with the stated goal of deriving an upper bound on allowable sensor noise for a desired pixel precision. The inequality should instead constrain the maximum pose covariance from above, e.g., $\lambda_{\max}\{\Sigma_u\} \leq \lambda_{\max}\{\Sigma_{\text{target}}\} \|J_{\xi}\|_2^2$. In addition, the notation $\sigma_{\max}$ is not clearly defined and appears to conflate the spectral norm of the Jacobian with that of the pose covariance matrix. The authors should clarify the notation and correct the bound accordingly.

In the annotation-efficiency comparison, the manual workflow is reported as 11.55 s/frame, while BirdsEye is reported as 3.27 s/frame. This directly corresponds to an approximately 3.5x improvement in frame-level annotation efficiency. However, the abstract and conclusion emphasize a 22–38x improvement in per-worker productivity. This larger factor appears to come from the number of generated labels or from multi-frame label reuse, but it is not directly comparable to conventional frame-by-frame annotation. The authors also acknowledge that this productivity estimate can be increased arbitrarily by raising the camera frame rate, which weakens its credibility as a measure of actual human labor efficiency. I recommend using more robust metrics, such as unique target instances per labor hour, effective training-sample quality per labor hour, or human time required to reach the same model performance.

The test set contains only one external site with 161 geotags, of which 141 are visible. The final recall is 83.7%, but the precision is only 35.1%. The authors argue that many false positives are actually unlabeled true targets, and therefore precision is not a reliable metric. However, this also indicates that the ground truth is incomplete and cannot reliably support the detection-performance evaluation. The authors should add a small but exhaustively annotated validation set, report standard precision/recall, F1 score, PR curves, or mAP, and compare against baselines such as manual-label training, training without soft labels, and different sigma settings.

The manuscript also lacks important implementation details such as the optimizer, learning rate, batch size, data split, threshold selection, and the rationale for DBSCAN parameter choices. These missing details weaken the reproducibility of the method.

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